Ex: $f(x) = x^3 + 3x^2 - 9x + 1$

increasing, decreasing, critical points, point of inflection, concavity, max, min

Critical points & Stationary points:

$$f'(x) = \underbrace{3x^2 + 6x - 9}$$

$$f'(x) = 0$$

$$\Rightarrow 3x^2 + 6x - 9 = 0$$

$$\Rightarrow 3(x^2 + 2x - 3) = 0$$

$$\Rightarrow x^2 + 3x - x - 3 = 0$$

$$\Rightarrow x(x+3) - 1(x+3) = 0$$

$$\Rightarrow (x+3)(x-1) = 0$$

$x = -3, 1 \rightarrow$ critical points / stationary points

Point of inflection

$$f''(x) = \underline{6x + 6}$$

$$\Rightarrow f''(x) = 0$$

$$\Rightarrow 6x + 6 = 0$$

$$\Rightarrow \boxed{x = -1}$$

$f'(x) > 0$

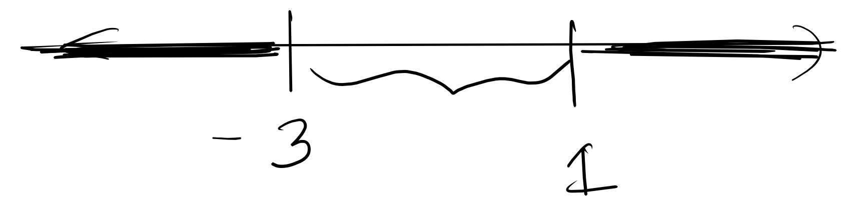
Increasing

$f'(x) < 0 \to$ Decreasing

$3(x+3)(x-1) < 0 \to$ Decreasing

$3x^2 + 6x - 9 > 0$

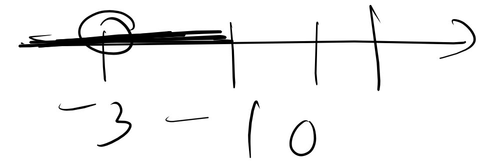
$\Rightarrow 3(x+3)(x-1) > 0 \to$ increasing



Interval	Test point	sign of $x+3$	sign of $x-1$	Sign of $f'(x)$	Decision
$x < -3$	-5	$-$	$-$	$+$	$f(x)$ is increasing on $(-\infty, -3)$.
$-3 < x < 1$	0	$+$	$-$	$-$	$f(x)$ is decreasing on $(-3, 1)$
$x > 1$	2	$+$	$+$	$+$	$f(x)$ is increasing on $(1, \infty)$.

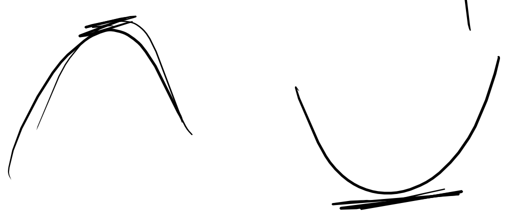
Concavity: $f''(x) > 0 \to$ concave up $f''(x) < 0$ concave down.

$f''(x) = 6(x+1)$



Interval	Test point	sign of $f''(x)$	Decision
$x < -1$	-2	$-$	$f(x)$ is concave down on $(-\infty, -1)$
$x > -1$	2	$+$	$f(x)$ is concave up on $(-1, \infty)$

$c = -3, 1$



$f''(-3) = 6(-3+1) = -12 < 0$, -3 is a maximum domain point

$f(-3) = (-3)^3 + 3(-3)^2 - 9(-3) + 1 = 28$ is maximum

$f''(1) = 12 > 0$, 1 is a minimum domain point.

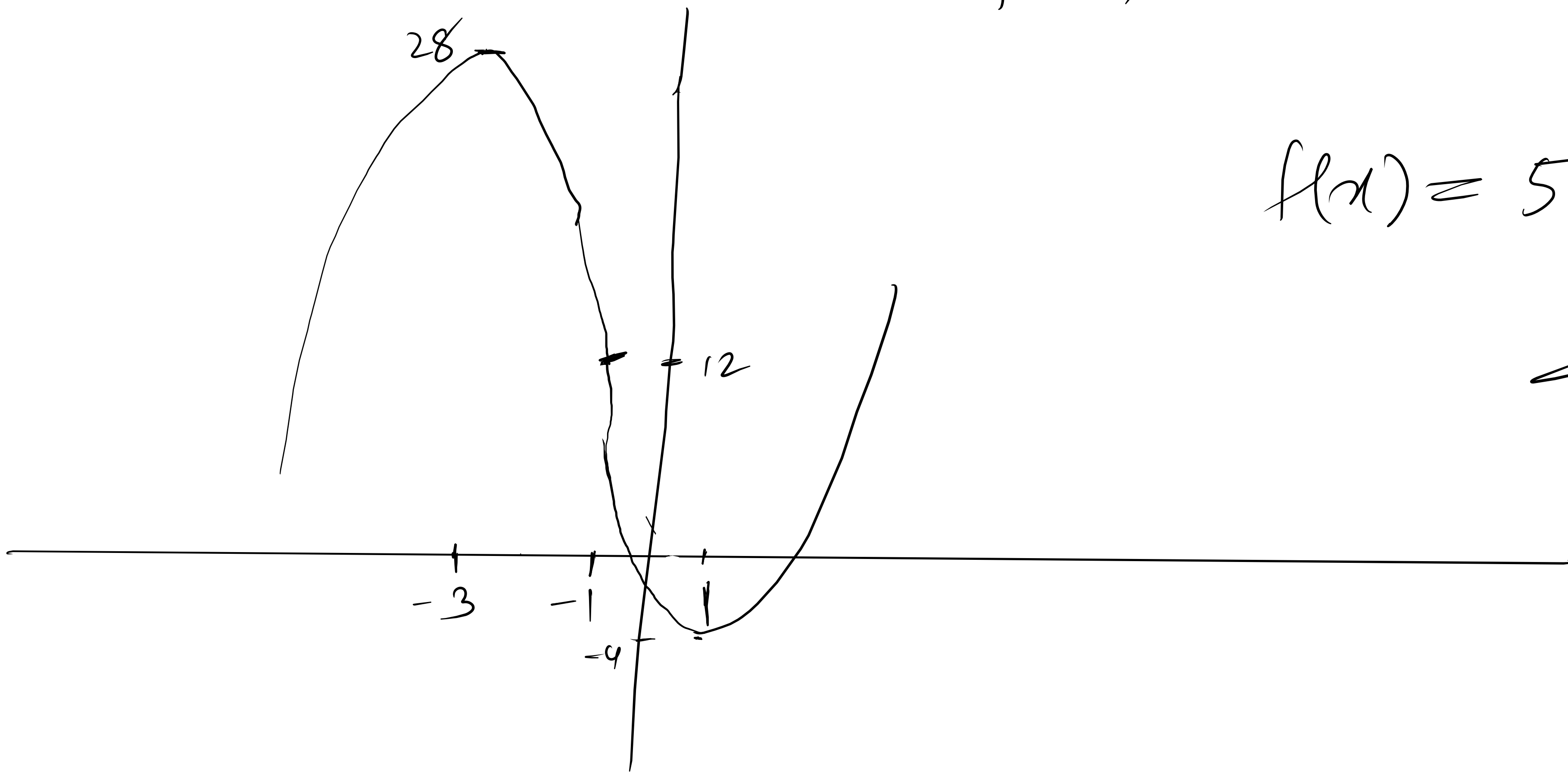
$f(1) = (1)^3 + 3(1)^2 - 9(1) + 1 = -4$ is a minimum point

Graph Sketching :

$-3 \to$ maximum domain , $f(-3) = 28$

$1 \to$ minimum domain , $f(1) = -4$

$-1 \to f(-1) = (-1)^3 + 3(-1)^2 - 9(-1) + 1 = 12$



$f(x) = 5 + 12x - x^3$

H.W.